

Stacked Intelligent Metasurfaces for Enhanced Secrecy in Integrated Sensing and Communication Systems

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Abstract—Abstract—Integrated Sensing and Communication (ISAC) systems are highly vulnerable to physical layer security threats. To address this, we propose a Stacked Intelligent Metasurfaces (SIM)-aided framework to enhance ISAC secrecy performance. Operating as a wave-domain precoder, the multi-layer SIM provides superior spatial degrees of freedom over conventional surfaces. We jointly optimize the active transmit beamforming and passive SIM phase shifts to maximize the worst-case secrecy rate (WSR) among legitimate users, subject to strict radar sensing beampattern, quality-of-service (QoS), and power constraints. To solve this severely non-convex problem, we develop an Alternating Optimization (AO) algorithm, employing Successive Convex Approximation (SCA) for the base station beamforming and a penalty-based gradient ascent method for the SIM phase shifts. Simulation results confirm that the proposed framework significantly outperforms baseline schemes, validating the substantial secrecy gains achieved by the multi-layer SIM architecture.

Index Terms—Stacked Intelligent Metasurfaces, Integrated Sensing and Communication, Physical Layer Security, Alternating Optimization.

I. INTRODUCTION

Integrated Sensing and Communication (ISAC) has emerged as a key technology for 6G networks by sharing spectrum and hardware between communication and radar functionalities [1]. However, illuminating sensing targets inherently exposes the system to severe eavesdropping threats [2]. To address this, Physical Layer Security (PLS) is increasingly utilized to degrade signal quality at unauthorized receivers by exploiting wireless channel characteristics [3]. While Reconfigurable Intelligent Surfaces (RIS) have been integrated into ISAC systems to proactively reshape the propagation environment for enhanced security [4], [5], their signal processing capability is fundamentally bottlenecked by a single interaction with incident electromagnetic waves.

To overcome this limitation, Stacked Intelligent Metasurfaces (SIM) have recently been introduced as a revolutionary multi-layer architecture [6]. By jointly optimizing phase shifts across cascaded metasurface layers, a SIM functions as an

advanced wave-domain precoder directly in the electromagnetic domain [7]. This design significantly increases the spatial degrees of freedom, offering unprecedented beamforming capabilities over conventional single-layer RIS.

While SIM shows tremendous potential, its application in secure ISAC systems remains largely unexplored due to the complex coupling between multi-layer phase shifts, transmit beamforming, and secrecy constraints. Motivated by this gap, this paper investigates the secrecy performance of a SIM-enabled ISAC system under perfect Channel State Information (CSI). Specifically, we propose a novel framework utilizing a multi-layer SIM to simultaneously enhance the communication secrecy rate and ensure a dedicated beampattern gain toward a radar target. To achieve this, we formulate a joint optimization problem to maximize the worst-case secrecy rate (WSR) among multiple users, subject to strict QoS, sensing power, and hardware constraints. Finally, to tackle the inherent non-convexity of the formulated problem, we develop an efficient Alternating Optimization (AO) algorithm that iteratively solves for the active transmit beamforming via Successive Convex Approximation (SCA) and optimizes the passive SIM phase shifts using a penalty-based gradient ascent method.

II. SECRECY PERFORMANCE UNDER PERFECT CSI

A. Downlink Communications Model

We consider a SIM-assisted ISAC system where a base station (BS) with N_t antennas serves K legitimate users (UEs) in the presence of K eavesdroppers (Eves). The SIM acts as a wave-domain precoder comprising L cascaded metasurface layers [6], each containing a uniform planar array (UPA) of M programmable meta-atoms. Let $\mathcal{L} = \{1, \dots, L\}$, $\mathcal{M} = \{1, \dots, M\}$, and $\mathcal{K} = \{1, \dots, K\}$ denote the index sets for the layers, meta-atoms, and users/eves, respectively.

The electromagnetic response of the l -th layer is defined by a diagonal matrix $\Phi_l = \text{diag}(e^{j\phi_1^l}, \dots, e^{j\phi_M^l}) \in \mathbb{C}^{M \times M}$, where $\phi_m^l \in [0, 2\pi)$ is the phase shift. Signal propagation between the $(l-1)$ -th and l -th layers is captured by the spatial transmission matrix $\Psi_l \in \mathbb{C}^{M \times M}$. Based on the Rayleigh-

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Sommerfeld diffraction theory [7], the (m, m') -th element of Ψ_l is formulated as

$$\psi_{m,m'}^l = \frac{A_0 \cos \theta_{m,m'}^l}{d_{m,m'}^l} \left(\frac{1}{2\pi d_{m,m'}^l} - j \frac{1}{\lambda} \right) \exp \left(j \frac{2\pi d_{m,m'}^l}{\lambda} \right), \quad (1)$$

where λ is the wavelength, $A_0 = dx \times dy$ is the meta-atom physical area, $d_{m,m'}^l$ is the transmission distance, and $\theta_{m,m'}^l$ is the elevation angle relative to the $(l-1)$ -th layer. Consequently, the equivalent forward wave-domain transmission matrix of the SIM, $\Theta \in \mathbb{C}^{M \times M}$, is given by

$$\Theta = \Phi_L \Psi_{L-1} \Phi_{L-1} \cdots \Psi_l \Phi_l \cdots \Psi_1 \Phi_1. \quad (2)$$

In the downlink, the BS transmits a precoded signal $\mathbf{s} = \sum_{k \in \mathcal{K}} \mathbf{w}_k s_k$, with normalized data symbols $s_k \sim \mathcal{CN}(0, 1)$ and active digital beamformers $\mathbf{w}_k \in \mathbb{C}^{N_t \times 1}$. The received signals at UE_k and Eve_k are respectively written as

$$y_k^{\text{UE}} = \mathbf{h}_k^H \Theta \mathbf{H} \sum_{i \in \mathcal{K}} \mathbf{w}_i s_i + n_k^{\text{UE}}, \quad (3)$$

$$y_k^{\text{Eve}} = \mathbf{v}_k^H \Theta \mathbf{H} \sum_{i \in \mathcal{K}} \mathbf{w}_i s_i + n_k^{\text{Eve}}, \quad (4)$$

where $n_k^{\text{UE}} \sim \mathcal{CN}(0, \sigma_k^2)$ and $n_k^{\text{Eve}} \sim \mathcal{CN}(0, \sigma_{e,k}^2)$ denote the AWGN. The baseband channel from the BS to the SIM is $\mathbf{H} \in \mathbb{C}^{M \times N_t}$. The channels from the SIM to UE_k and Eve_k, denoted as $\mathbf{h}_k^H$ and $\mathbf{v}_k^H$, are modeled as correlated Rayleigh fading [8]:

$$\mathbf{h}_k, \mathbf{v}_k \sim \mathcal{CN}(\mathbf{0}, \alpha_k \mathbf{R}), \quad \forall k \in \mathcal{K}. \quad (5)$$

Here, $\alpha_k = C_0(d_k/d_0)^{-\bar{n}}$ characterizes the distance-dependent path loss, with $C_0 = (\lambda/4\pi d_0)^2$ representing the reference path loss at $d_0 = 1$ m. Assuming an isotropic scattering environment [8], the spatial correlation matrix $\mathbf{R} \in \mathbb{C}^{M \times M}$ among the meta-atoms is defined by

$$\mathbf{R}_{m,m'} = \text{sinc} \left(\frac{2d_{m,m'}}{\lambda} \right), \quad (6)$$

where $d_{m,m'}$ is the physical element spacing.

B. Sensing Model

For sensing, the SIM's output layer operates as a UPA with half-wavelength $(\lambda/2)$ spacing. It synthesizes a highly directional beam toward a far-field target located at elevation $\theta \in (0, \pi)$ and azimuth $\phi \in (-\pi/2, \pi/2)$. With M_x and M_z meta-atoms along the x - and z -axes ($M = M_x M_z$), the normalized spatial steering vector $\mathbf{a}(\theta, \phi) \in \mathbb{C}^{M \times 1}$ is formulated as [1]:

$$\mathbf{a}(\theta, \phi) = \frac{1}{\sqrt{M_x M_z}} \mathbf{a}_x(\theta, \phi) \otimes \mathbf{a}_z(\theta), \quad (7)$$

where the 1D array response vectors are defined as

$$\mathbf{a}_x(\theta, \phi) = [1, e^{-j\pi \sin \theta \sin \phi}, \dots, e^{-j(M_x-1)\pi \sin \theta \sin \phi}]^T \in \mathbb{C}^{M_x \times 1} \quad (8)$$

$$\mathbf{a}_z(\theta) = [1, e^{-j\pi \cos \theta}, \dots, e^{-j(M_z-1)\pi \cos \theta}]^T \in \mathbb{C}^{M_z \times 1}. \quad (9)$$

C. Performance Metrics

1) *Communication Metric*: Based on (3) and (4), the achievable rates (in bpcu) for legitimate users and eavesdroppers are

$$R_k^{\text{UE}} = \log_2(1 + \gamma_k^{\text{UE}}), \quad (10)$$

$$R_k^{\text{Eve}} = \log_2(1 + \gamma_k^{\text{Eve}}), \quad (11)$$

where the respective signal-to-interference-plus-noise ratios (SINRs) are expanded as

$$\gamma_k^{\text{UE}} = \frac{|\mathbf{h}_k^H \Theta \mathbf{H} \mathbf{w}_k|^2}{\sum_{i \neq k} |\mathbf{h}_k^H \Theta \mathbf{H} \mathbf{w}_i|^2 + \sigma_k^2}, \quad (12)$$

$$\gamma_k^{\text{Eve}} = \frac{|\mathbf{v}_k^H \Theta \mathbf{H} \mathbf{w}_k|^2}{\sum_{i \neq k} |\mathbf{v}_k^H \Theta \mathbf{H} \mathbf{w}_i|^2 + \sigma_{e,k}^2}. \quad (13)$$

The system's worst-case secrecy rate (WSR) among all users is defined as

$$\Delta R_{\min} = \min_{k \in \mathcal{K}} [R_k^{\text{UE}} - R_k^{\text{Eve}}]^+, \quad (14)$$

where $[a]^+ \triangleq \max\{a, 0\}$.

2) *Sensing Metric*: The transmit beampattern gain towards the target location (θ_c, ϕ_c) is derived by aggregating the power from all communication beams:

$$P(\theta_c, \phi_c) = \sum_{k \in \mathcal{K}} |\mathbf{a}^H(\theta_c, \phi_c) \Theta \mathbf{H} \mathbf{w}_k|^2. \quad (15)$$

Reliable target detection requires the condition $P(\theta_c, \phi_c) \geq \Gamma$, where Γ is the minimum sensing power threshold.

D. Problem Formulation

We aim to jointly optimize the BS digital beamforming $\{\mathbf{w}_k\}$ and the SIM wave-domain phase shifts $\{\Phi_l\}$ to maximize the WSR, subject to sensing and hardware constraints:

$$\max_{\{\mathbf{w}_k\}, \{\Phi_l\}} \Delta R_{\min} \quad (16a)$$

$$\text{s.t. } R_k^{\text{UE}} \geq R_{\text{th}}, \quad \forall k \in \mathcal{K}, \quad (16b)$$

$$P(\theta_c, \phi_c) \geq \Gamma, \quad (16c)$$

$$\sum_{k \in \mathcal{K}} \|\mathbf{w}_k\|^2 \leq P_{\max}, \quad (16d)$$

$$\phi_m^l \in [0, 2\pi), \quad \forall m \in \mathcal{M}, \forall l \in \mathcal{L}. \quad (16e)$$

In (16), constraint (16b) ensures a minimum QoS threshold R_{th} for legitimate users, (16c) guarantees the required sensing power Γ , (16d) bounds the BS active transmit power by $P_{\max}$, and (16e) imposes the passive unit-modulus limitation on the SIM meta-atoms.

III. PROPOSED OPTIMIZATION ALGORITHM

In this section, we develop a computationally efficient algorithm to tackle the secrecy-oriented joint design problem formulated in (16). The core challenge lies in the joint optimization of the BS active transmit beamforming vectors $\{\mathbf{w}_k\}$ and the multi-layer SIM passive phase shift matrices $\{\Phi_l\}$. Because these variables are tightly coupled across the objective function and the constraints, the problem cannot be solved for a global optimum in a closed form.

Specifically, the severe non-convexity of problem (16) originates from three main aspects:

- 1) **Non-concave Objective Function:** The secrecy capacity is defined as the difference between two logarithmic functions:

$$R_k^{\text{sec}} = \log_2(1 + \gamma_k^{\text{UE}}) - \log_2(1 + \gamma_k^{\text{Eve}}). \quad (17)$$

The difference between two concave functions is generally neither convex nor concave, posing a strict non-concave maximization problem.

- 2) **Bilinear Variable Coupling:** The SINR expressions involve quadratic fractional terms of the form

$$|\mathbf{h}_k^H \Theta \mathbf{H} \mathbf{w}_k|^2. \quad (18)$$

Given that Θ is a product of $\{\Phi_l\}$, the digital beamformers $\{\mathbf{w}_k\}$ and the SIM phase variables are bilinearly coupled, rendering the constraints non-convex.

- 3) **Unit-Modulus Constraints:** The passive nature of the SIM imposes strict hardware constraints on the phase shifts, mathematically represented by the unit-modulus constraint $|e^{j\phi_m}| = 1$, which forms a highly non-convex feasible set.

To circumvent these mathematical hurdles, we propose an alternating optimization (AO) framework that strategically decomposes the original complex problem into two more tractable subproblems.

A. AO Framework Formulation

Let $\mathbf{W} \triangleq [\mathbf{w}_1, \dots, \mathbf{w}_K] \in \mathbb{C}^{N_t \times K}$ denote the composite digital beamforming matrix. The interference-plus-noise powers at the k -th legitimate user and eavesdropper are respectively defined as

$$I_k^{\text{UE}}(\mathbf{W}, \{\Phi_l\}) = \sum_{j \neq k} |\mathbf{h}_k^H \Theta \mathbf{H} \mathbf{w}_j|^2 + \sigma_k^2, \quad (19)$$

$$I_k^{\text{Eve}}(\mathbf{W}, \{\Phi_l\}) = \sum_{j \neq k} |\mathbf{v}_k^H \Theta \mathbf{H} \mathbf{w}_j|^2 + \sigma_{e,k}^2. \quad (20)$$

To handle the non-smooth minimum operator in $\Delta R_{\min}$, we introduce a slack continuous variable t representing the worst-case secrecy rate. By applying the epigraph reformulation technique, the original problem is equivalently transformed into

$$\max_{t, \mathbf{W}, \{\Phi_l\}} t \quad (21a)$$

$$\text{s.t. } R_k^{\text{UE}} - R_k^{\text{Eve}} \geq t, \quad \forall k \in \mathcal{K}, \quad (21b)$$

$$(16b), (16c), (16d), (16e). \quad (21c)$$

B. Beamforming Optimization via SCA

For fixed $\{\Phi_l\}$, optimizing $\mathbf{W}$ remains non-convex. We employ the successive convex approximation (SCA) technique [9] to construct tractable surrogate functions at a local point $\mathbf{W}^{(i)}$. Using first-order Taylor expansions and inequality approximations [10], the legitimate user's rate is lower-bounded by

$$R_k^{\text{UE}}(\mathbf{W}) \geq \tilde{R}_k^{\text{low},(i)}(\mathbf{W}) \triangleq a_k^{(i)} + b_k^{(i)} \left(2 - \frac{|\mathbf{h}_k^H \Theta \mathbf{H} \mathbf{w}_k^{(i)}|^2}{\mathcal{L}_k^{(i)}(\mathbf{W})} - \frac{I_k^{\text{UE}}(\mathbf{W})}{I_k^{\text{UE},(i)}(\mathbf{W}^{(i)})} \right), \quad (22)$$

where the trust region ensuring a positive denominator is defined as

$$\mathcal{L}_k^{(i)}(\mathbf{W}) \triangleq 2 \operatorname{Re} \left\{ (\mathbf{h}_k^H \Theta \mathbf{H} \mathbf{w}_k^{(i)})^* (\mathbf{h}_k^H \Theta \mathbf{H} \mathbf{w}_k) \right\} - |\mathbf{h}_k^H \Theta \mathbf{H} \mathbf{w}_k^{(i)}|^2 \geq 0. \quad (23)$$

The constant coefficients evaluated at the i -th iteration are

$$a_k^{(i)} = \log_2 \left(1 + \frac{|\mathbf{h}_k^H \Theta \mathbf{H} \mathbf{w}_k^{(i)}|^2}{I_k^{\text{UE},(i)}(\mathbf{W}^{(i)})} \right),$$

$$b_k^{(i)} = \frac{|\mathbf{h}_k^H \Theta \mathbf{H} \mathbf{w}_k^{(i)}|^2 / I_k^{\text{UE},(i)}(\mathbf{W}^{(i)})}{\ln 2 \left(1 + |\mathbf{h}_k^H \Theta \mathbf{H} \mathbf{w}_k^{(i)}|^2 / I_k^{\text{UE},(i)}(\mathbf{W}^{(i)}) \right)}.$$

Similarly, Eve's rate is upper-bounded by the linear inequality [10]:

$$R_k^{\text{Eve}}(\mathbf{W}) \leq \tilde{R}_k^{\text{up,Eve},(i)}(\mathbf{W}) \triangleq \log_2 \left(1 + \frac{|\mathbf{v}_k^H \Theta \mathbf{H} \mathbf{w}_k^{(i)}|^2}{I_k^{\text{Eve},(i)}(\mathbf{W}^{(i)})} \right) + \frac{1}{\ln 2} \left(1 + \frac{|\mathbf{v}_k^H \Theta \mathbf{H} \mathbf{w}_k^{(i)}|^2}{I_k^{\text{Eve},(i)}(\mathbf{W}^{(i)})} \right)^{-1} \times \left(\frac{|\mathbf{v}_k^H \Theta \mathbf{H} \mathbf{w}_k|^2}{\mathcal{L}_k^{\text{Eve},(i)}(\mathbf{W}) + \sigma_{e,k}^2} - \frac{|\mathbf{v}_k^H \Theta \mathbf{H} \mathbf{w}_k^{(i)}|^2}{I_k^{\text{Eve},(i)}(\mathbf{W}^{(i)})} \right), \quad (24)$$

with the associated trust region given by

$$\mathcal{L}_k^{\text{Eve},(i)}(\mathbf{W}) \triangleq 2 \operatorname{Re} \left\{ \sum_{j \neq k} (\mathbf{v}_k^H \Theta \mathbf{H} \mathbf{w}_j^{(i)})^* (\mathbf{v}_k^H \Theta \mathbf{H} \mathbf{w}_j) \right\} - \sum_{j \neq k} |\mathbf{v}_k^H \Theta \mathbf{H} \mathbf{w}_j^{(i)}|^2 \geq 0. \quad (25)$$

Consequently, the secrecy constraint is conservatively approximated by

$$\tilde{R}_k^{\text{low},(i)}(\mathbf{W}) - \tilde{R}_k^{\text{up,Eve},(i)}(\mathbf{W}) \geq t, \quad \forall k \in \mathcal{K}. \quad (26)$$

Applying the first-order Taylor expansion, the radar sensing constraint is lower-bounded by

$$P(\mathbf{W}) \geq P^{\text{low},(i)}(\mathbf{W}), \quad (27)$$

where the linear surrogate function is

$$P^{\text{low},(i)}(\mathbf{W}) \triangleq 2 \operatorname{Re} \left\{ \sum_{k \in \mathcal{K}} (\mathbf{a}^H \mathbf{\Theta} \mathbf{H} \mathbf{w}_k^{(i)})^* (\mathbf{a}^H \mathbf{\Theta} \mathbf{H} \mathbf{w}_k) \right\} - \sum_{k \in \mathcal{K}} |\mathbf{a}^H \mathbf{\Theta} \mathbf{H} \mathbf{w}_k^{(i)}|^2. \quad (28)$$

Replacing the non-convex components with their convex surrogates yields the convexified beamforming subproblem at the i -th iteration:

$$\max_{\mathbf{W}, t} \quad t \quad (29a)$$

$$\text{s.t.} \quad \tilde{R}_k^{\text{low},(i)}(\mathbf{W}) - \tilde{R}_k^{\text{up,Eve},(i)}(\mathbf{W}) \geq t, \quad \forall k \in \mathcal{K}, \quad (29b)$$

$$P^{\text{low},(i)}(\mathbf{W}) \geq \Gamma, \quad (29c)$$

$$\sum_{k \in \mathcal{K}} \|\mathbf{w}_k\|^2 \leq P_{\max}, \quad (29d)$$

$$\mathcal{L}_k^{(i)}(\mathbf{W}) \geq 0, \quad \mathcal{L}_k^{\text{Eve},(i)}(\mathbf{W}) \geq 0, \quad \forall k \in \mathcal{K}. \quad (29e)$$

This subproblem constitutes a standard convex optimization problem and can be efficiently solved using standard off-the-shelf convex solvers such as CVX.

C. SIM Phase Optimization with Fixed Beamforming

For a fixed active beamforming matrix $\mathbf{W}$, we optimize the SIM phase shifts $\{\phi_m^l\}$ via a gradient ascent algorithm. To effectively handle the sensing constraint, we incorporate it as a penalty term within the objective function [11]:

$$\max_{\{\phi_m^l\}} \quad F = \Delta R_{\min} + \beta g_c, \quad (30)$$

$$\text{s.t.} \quad \Delta R_{\min} = \min_{k \in \mathcal{K}} (R_k^{\text{UE}} - R_k^{\text{Eve}}), \quad (31)$$

$$g_c = \min \{P(\theta_c, \phi_c) - \Gamma, 0\}, \quad (32)$$

$$0 \leq \phi_m^l < 2\pi, \quad \forall m \in \mathcal{M}, \forall l \in \mathcal{L}, \quad (33)$$

where $\beta > 0$ is the penalty factor. By applying the multi-variable chain rule, the gradient is expressed as

$$\frac{\partial F}{\partial \phi_m^l} = \frac{\partial \Delta R_{\min}}{\partial \phi_m^l} + \beta \frac{\partial g_c}{\partial \phi_m^l}. \quad (34)$$

For the user k^* achieving the minimum secrecy rate, we have $\frac{\partial \Delta R_{\min}}{\partial \phi_m^l} = \frac{\partial R_{k^*}^{\text{UE}}}{\partial \phi_m^l} - \frac{\partial R_{k^*}^{\text{Eve}}}{\partial \phi_m^l}$. The legitimate rate derivative is derived as

$$\frac{\partial R_k^{\text{UE}}}{\partial \phi_m^l} = \frac{1}{\ln 2 (1 + \gamma_k^{\text{UE}})} \left[\frac{1}{I_k^{\text{UE}}} \frac{\partial \delta_{k,k}^{\text{UE}}}{\partial \phi_m^l} - \frac{(\gamma_k^{\text{UE}})^2}{\delta_{k,k}^{\text{UE}}} \sum_{j \neq k} \frac{\partial \delta_{k,j}^{\text{UE}}}{\partial \phi_m^l} \right], \quad (35)$$

where $\delta_{k,j}^{\text{UE}} \triangleq |\mathbf{h}_k^H \mathbf{\Theta} \mathbf{H} \mathbf{w}_j|^2$. By decomposing the transmission matrix [11], the derivative of the signal power component $\delta_{k,j}^{\text{UE}}$ is explicitly calculated as

$$\frac{\partial \delta_{k,j}^{\text{UE}}}{\partial \phi_m^l} = 2 \operatorname{Re} \left\{ j e^{j \phi_m^l} [\mathbf{h}_k^H \mathbf{L}_l]_m [\mathbf{K}_l \mathbf{w}_j]_m (\mathbf{h}_k^H \mathbf{\Theta} \mathbf{H} \mathbf{w}_j)^* \right\}, \quad (36)$$

where the forward matrix is $\mathbf{K}_1 = \mathbf{H}$ and $\mathbf{K}_l = \mathbf{\Psi}_{l-1} \mathbf{\Phi}_{l-1} \cdots \mathbf{\Psi}_1 \mathbf{\Phi}_1 \mathbf{H}$ for $l > 1$. Symmetrically, the

backward matrix is defined as $\mathbf{L}_L = \mathbf{I}_M$ and $\mathbf{L}_l = \mathbf{\Phi}_L \mathbf{\Psi}_{L-1} \cdots \mathbf{\Phi}_{l+1} \mathbf{\Psi}_l$ for $l < L$.

Following a similar derivation pattern, Eve's rate derivative is

$$\frac{\partial R_k^{\text{Eve}}}{\partial \phi_m^l} = \frac{1}{\ln 2 (1 + \gamma_k^{\text{Eve}})} \left[\frac{1}{I_k^{\text{Eve}}} \frac{\partial \delta_{k,k}^{\text{Eve}}}{\partial \phi_m^l} - \frac{(\gamma_k^{\text{Eve}})^2}{\delta_{k,k}^{\text{Eve}}} \sum_{j \neq k} \frac{\partial \delta_{k,j}^{\text{Eve}}}{\partial \phi_m^l} \right], \quad (37)$$

with the leaked power derivative given by

$$\frac{\partial \delta_{k,j}^{\text{Eve}}}{\partial \phi_m^l} = 2 \operatorname{Re} \left\{ j e^{j \phi_m^l} [\mathbf{v}_k^H \mathbf{L}_l]_m [\mathbf{K}_l \mathbf{w}_j]_m (\mathbf{v}_k^H \mathbf{\Theta} \mathbf{H} \mathbf{w}_j)^* \right\}. \quad (38)$$

The gradient of the sensing penalty term is conditionally evaluated as $\frac{\partial P(\theta_c, \phi_c)}{\partial \phi_m^l}$ if $P(\theta_c, \phi_c) < \Gamma$, and 0 otherwise, where

$$\frac{\partial P(\theta_c, \phi_c)}{\partial \phi_m^l} = \sum_{k \in \mathcal{K}} 2 \operatorname{Re} \left\{ j e^{j \phi_m^l} [\mathbf{a}^H \mathbf{L}_l]_m [\mathbf{K}_l \mathbf{w}_k]_m (\mathbf{a}^H \mathbf{\Theta} \mathbf{H} \mathbf{w}_k)^* \right\}. \quad (39)$$

To ensure numerical stability and mitigate severe oscillations, the gradients are normalized at each iteration as

$$\frac{\partial F}{\partial \phi_m^l} \leftarrow \frac{\pi}{\eta} \frac{\partial F}{\partial \phi_m^l}, \quad \text{where } \eta = \max_{m,l} \left| \frac{\partial F}{\partial \phi_m^l} \right|. \quad (40)$$

Finally, the SIM phase shifts are updated dynamically using the Armijo backtracking step size $\mu > 0$:

$$\phi_m^l \leftarrow \phi_m^l + \mu \frac{\partial F}{\partial \phi_m^l}, \quad \forall m \in \mathcal{M}, \forall l \in \mathcal{L}. \quad (41)$$

IV. SIMULATION RESULTS

In this section, we evaluate the secrecy performance of the proposed SIM-enabled ISAC system through numerical simulations. Unless otherwise specified, the system parameters are set as follows: the BS is equipped with $N_t = 8$ antennas, serving $K = 2$ legitimate users in the presence of $K = 2$ eavesdroppers. The SIM consists of $L = 3$ layers, with each layer containing $M = 100$ (10×10) meta-atoms. The maximum transmit power at the BS is $P_{\max} = 30$ dBm, and the noise power at all receivers is set to $\sigma_u^2 = \sigma_e^2 = -90$ dBm. The radar sensing target is located at the elevation and azimuth angles of $(\theta_c, \phi_c) = (30^\circ, 45^\circ)$, with a predefined sensing threshold of $\Gamma = -10$ dBm.

A. Convergence Analysis

Fig. 1 illustrates the convergence behavior of the proposed Alternating Optimization (AO) algorithm. It is observed that the worst-case secrecy rate (WSR) monotonically increases and stabilizes within approximately 8 to 10 iterations, which verifies the efficiency and convergence guarantee of the proposed SCA and penalty-based Gradient Ascent (SIM-GA) methods. Furthermore, the proposed SIM-GA scheme achieves a substantial performance gain (over 14 bps/Hz) compared to the baseline schemes.

Noticeably, the baseline scheme with random SIM phase shifts yields worse secrecy performance compared to the

conventional digital beamforming system without phase optimization (i.e., utilizing SCA only). This phenomenon is physically intuitive. In our considered architecture, the multi-layer SIM acts as a cascaded wave-domain precoder directly integrated at the transmitter. Consequently, assigning random phase shifts across multiple passive layers severely scatters the electromagnetic waves and causes destructive interference. This severely degrades the carefully designed digital beam-pattern and the overall system performance, highlighting the critical necessity of jointly optimizing the multi-layer wave-domain precoder.

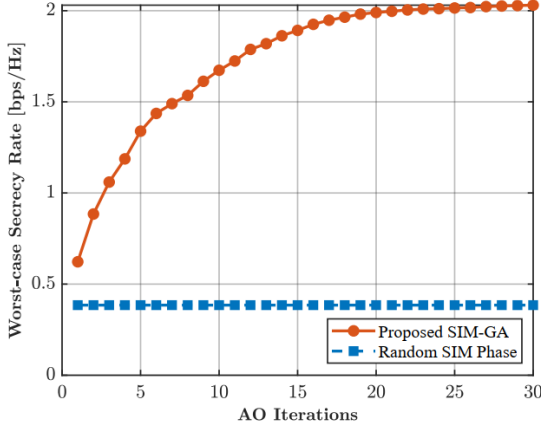


Fig. 1: Worst-case secrecy rate versus the number of AO iterations.

B. Impact of the Number of SIM Layers

Fig. 2 depicts the average WSR versus the number of cascaded SIM layers L . The system performance exhibits a general upward trend as L increases from 1 to 4, reaching a peak at $L = 4$. This improvement stems from the enhanced spatial degrees of freedom provided by the additional metasurface layers, which empower the system with more profound wave-manipulation capabilities to mitigate inter-user interference and degrade eavesdropper channels. However, the WSR shows fluctuations and a slight degradation for $L > 4$. This phenomenon can be attributed to the increased optimization complexity of the highly non-convex landscape, making it challenging for the gradient ascent algorithm to escape local optima within limited iterations, as well as the cumulative signal attenuation caused by propagation through excessive passive layers.

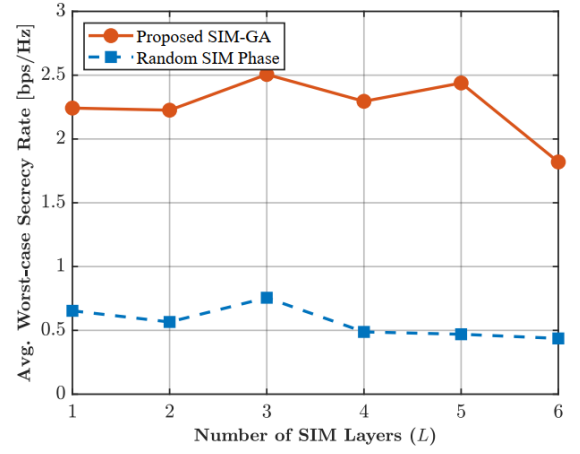


Fig. 2: Average worst-case secrecy rate versus the number of SIM layers (L).

C. Impact of Maximum Transmit Power

Fig. 3 demonstrates the impact of the BS maximum transmit power budget, P_{max} , on the network secrecy performance. As expected, the WSR increases monotonically with P_{max} , growing from approximately 16.5 bps/Hz at 20 dBm to over 22.5 bps/Hz at 40 dBm. This trend confirms that the proposed algorithm efficiently allocates the available power budget. By smartly reconfiguring the wave-domain propagation via the SIM, the system effectively utilizes the extra transmit power to boost the legitimate users' capacity and fulfill the radar sensing requirement without inadvertently enhancing the eavesdroppers' interception capabilities.

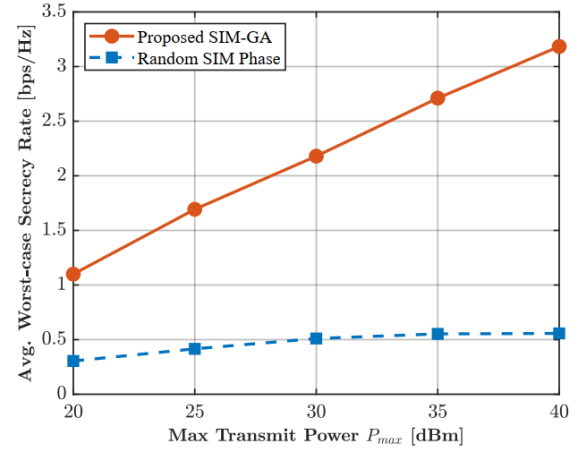


Fig. 3: Average worst-case secrecy rate versus the maximum transmit power (P_{max}).

D. Trade-off between Sensing and Secure Communication

Fig. 4 illustrates the fundamental trade-off between the radar sensing capability and the communication secrecy performance by plotting the average WSR against the sensing beampattern gain threshold Γ . It can be clearly observed that the WSR gradually degrades as the sensing requirement becomes more stringent (i.e., as Γ increases from -20 dBm to 10 dBm). This behavior is theoretically expected in ISAC

systems: forcing the system to guarantee a higher power focus towards the radar target consumes a significant portion of the spatial degrees of freedom and transmit power. Consequently, fewer resources are available for the base station and the SIM to optimize the legitimate users' channels and effectively nullify the signal leakage toward the eavesdroppers. Nevertheless, the proposed multi-layer SIM framework maintains a robust secrecy rate even under demanding sensing conditions, demonstrating its flexibility in balancing dual functions.

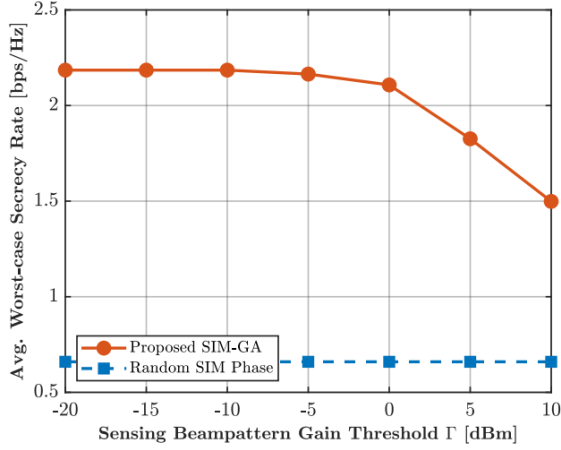


Fig. 4: Average worst-case secrecy rate versus the sensing beampattern gain threshold (Γ).

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